Scientific English–Hindi Lexicon & Teaching Reference

A Complete 250-Item Guide for Classroom Discourse, Lab Instruction, and Academic Writing

PART 1: CORE SCIENTIFIC NOUNS (ITEMS 1–25)

Nouns: Set A

#	Term (Noun)	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
1	Hypothesis	परिकल्पना (Parikalpana)	A testable proposed explanation for an observed event.	*"State your hypothesis clearly before beginning the lab trial."*
2	Apparatus	उपकरण (Upakaran)	Equipment or assembly needed to conduct an experiment.	*"Check the glass joints of the distillation apparatus for leaks."*
3	Meniscus	नवचंद्रक (Navachandrak)	The curved upper surface of liquid in a graduated tube.	*"Read the liquid volume strictly at the bottom of the meniscus."*
4	Precipitate	अवक्षेप (Avakshep)	Insoluble solid separating out from a liquid solution.	*"A yellow precipitate of lead iodide settles to the flask base."*
5	Catalyst	उत्प्रेरक (Utprerak)	Substance increasing reaction speed without being consumed.	*"Manganese dioxide acts as an active catalyst in this trial."*
6	Solute	विलेय (Viley)	The substance dissolved within a solvent.	*"Table salt acts as the solute when dissolved into warm water."*
7	Solvent	विलायक (Vilaayak)	Liquid liquid medium that dissolves another substance.	*"Ethanol functions as an organic solvent in leaf pigment extraction."*
8	Substrate	क्रियाधार (Kriyaadhaar)	Surface or chemical reactant acted on by an enzyme.	*"The enzyme binds specifically to the active site on the substrate."*
9	Equilibrium	साम्यावस्था (Saamyaavastha)	Balanced state where opposing processes occur equally.	*"Dynamic equilibrium is reached when forward and reverse rates match."*
10	Osmosis	परासरण (Paraasaran)	Solvent diffusion across semipermeable membrane.	*"Water enters plant root hairs primarily through osmosis."*
11	Viscosity	श्यानता (Shyaanata)	Measure of a fluid's internal resistance to flowing.	*"Glycerin exhibits significantly higher viscosity than water."*
12	Titration	अनुमापन (Anumaapan)	Method using standard titrant to find solution strength.	*"Carry out the acid-base titration dropwise until endpoint color shifts."*
13	Specimen	नमूना / प्रतिदर्श (Namuna)	Sample of organism, tissue, or mineral for examination.	*"Place the botanical specimen in the center of the clean glass slide."*
14	Entropy	एन्ट्रॉपी / अव्यवस्था (Entropy)	Thermodynamic measure of molecular disorder.	*"The entropy of an isolated thermodynamic system always increases."*
15	Enthalpy	पूर्ण ऊष्मा / एन्थैल्पी (Enthalpy)	Total internal heat content under constant pressure.	*"Calculate the standard enthalpy of combustion for methane gas."*
16	Homeostasis	समस्थापन (Samsthaapan)	Self-regulating balance of internal physiological conditions.	*"Thermoregulation is a critical homeostatic loop in mammals."*
17	Isotope	समस्थानिक (Samsthaanik)	Atoms with same atomic number but different mass numbers.	*"Carbon-14 is an unstable radioactive isotope used in dating."*
18	Photosynthesis	प्रकाश संश्लेषण (Prakash Sanshleshan)	Process using sunlight to synthesize glucose from CO2.	*"Chlorophyll traps light photons to power photosynthesis."*
19	Diffraction	विवर्तन (Vivartan)	Bending of waves encountering an obstacle or narrow slit.	*"Laser light shows clear diffraction patterns on the classroom screen."*
20	Refraction	अपवर्तन (Apvartan)	Change in wave direction passing between two media.	*"Refraction causes the pencil to appear bent when submerged in water."*
21	Inertia	जड़त्व (Jadatva)	Resistance of a body to any change in state of motion.	*"Newton's first law formalizes the physical concept of inertia."*
22	Momentum	संवेग (Samveg)	Product of mass and velocity of a moving object.	*"Total linear momentum remains conserved during an elastic collision."*
23	Friction	घर्षण (Gharshan)	Opposing resistive force between contacting sliding surfaces.	*"Apply lubricant to minimize frictional wear on moving joints."*
24	Centrifuge	अपकेंद्रित्र (Apkendritra)	Device spinning tubes rapidly to separate densities.	*"Spin the blood vial in the centrifuge for five minutes at 3000 RPM."*
25	Petri dish	पेट्री डिश (Petri Dish)	Shallow transparent lidded dish for culturing microbes.	*"Inoculate agar in the petri dish using a sterile loop."*

PART 1: SCIENTIFIC NOUNS CONTINUED (ITEMS 26–50)

Nouns: Set B

#	Term (Noun)	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
26	Pipette	पिपेट (Pipette)	Slender calibrated tube for transferring exact liquid aliquots.	*"Use a micropipette to transfer ten microliters of reagent."*
27	Burette	ब्यूरेट (Burette)	Graduated vertical tube with stopcock for titrations.	*"Fill the burette to the zero mark and ensure no air bubbles stay trapped."*
28	Control group	नियंत्रित समूह (Niyantrit Samooh)	Experimental baseline kept free from the tested treatment.	*"Compare seedlings in the treated group against the control group."*
29	Empiricism	अनुभववाद / प्रयोगवाद (Anubhavvaad)	Theory that knowledge derives from sensory observation.	*"Modern scientific inquiry relies firmly on empiricism and data."*
30	Biodiversity	जैव विविधता (Jaiv Vividhta)	Variety of life forms within a given habitat or biosphere.	*"Habitat destruction causes a measurable drop in local biodiversity."*
31	Mitochondria	माइटोकॉन्ड्रिया (Mitochondria)	Cell organelles generating ATP through cellular respiration.	*"Mitochondria are often referred to as the powerhouse of the cell."*
32	Reagent	अभिकर्मक (Abhikarmak)	Chemical compound added to test or drive a reaction.	*"Store the oxidizing reagent inside an amber glass bottle."*
33	Electrolysis	विद्युत अपघटन (Vidyut Apgathan)	Decomposition of chemical compounds via electrical current.	*"Water splits into hydrogen and oxygen through electrolysis."*
34	Combustion	दहन (Dahan)	Rapid exothermic chemical reaction with oxygen.	*"Complete combustion of propane releases water and carbon dioxide."*
35	Resonance	अनुनाद (Anunaad)	Large amplitude oscillation matching natural frequency.	*"The tuning fork vibrates loudly when acoustic resonance is met."*
36	Superconductivity	अतिचालकता (Atichalakata)	Zero electrical resistance below critical temperature.	*"Certain ceramics exhibit superconductivity when cooled with liquid nitrogen."*
37	Chromosome	गुणसूत्र (Gunasootra)	Thread-like DNA structure carrying genetic information.	*"Human somatic cells contain twenty-three pairs of chromosomes."*
38	Mutation	उत्परिवर्तन (Utparivartan)	Alteration in nucleotide sequence of genetic material.	*"Ultraviolet exposure can induce mutations in bacterial cultures."*
39	Sediment	तलछट / गाद (Talchhat)	Particulate mineral matter deposited by water, wind, or ice.	*"Allow coarse sand and silt sediment to settle before decanting."*
40	Ecosystem	पारिस्थितिकी तंत्र (Paaristhitiki)	Community of interacting living organisms and physical abiotic environment.	*"Mangrove forests form a delicate coastal ecosystem."*
41	Wavelength	तरंगदैर्ध्य (Tarangdairdhya)	Distance between consecutive corresponding wave crests.	*"Red visible light features a longer wavelength than blue light."*
42	Frequency	आवृति (Aavriti)	Number of complete wave cycles passing a point per second.	*"Frequency is measured in Hertz and relates inversely to wavelength."*
43	Capillarity	केशिकात्व (Keshikatva)	Tendency of liquids to rise or fall in narrow tubes.	*"Capillarity allows sap to travel upwards through plant xylem tubes."*
44	Titer	अनुमापांक (Anumaapaank)	Concentration of an active substance determined by titration.	*"Record the exact final titer to compute normality of the acid."*
45	Distillate	आसूत (Aasoot)	Vapor condensed into purified liquid during distillation.	*"Collect the volatile distillate in a chilled receiver flask."*
46	Thermal conductivity	ऊष्मा चालकता (Ooshma Chalakta)	Property of a material quantifying heat transfer rate.	*"Copper possesses high thermal conductivity compared to glass."*
47	Specific heat	विशिष्ट ऊष्मा (Vishisht Ooshma)	Heat required to raise temperature of unit mass by 1°C.	*"Water's high specific heat buffers temperature fluctuations near coasts."*
48	Oxidation	ऑक्सीकरण / उपचयन (Upchayan)	Loss of electrons or increase in chemical oxidation state.	*"Rusting of iron nails is a gradual atmospheric oxidation process."*
49	Reduction	अपचयन (Apchayan)	Gain of electrons or decrease in chemical oxidation state.	*"At the cathode terminal, copper ions undergo reduction to solid metal."*
50	Combustion chamber	दहन कक्ष (Dahan Kaksh)	Enclosed space where fuel fuel-air mixture ignites.	*"Maintain constant air-to-fuel ratio inside the combustion chamber."*

PART 2: SCIENTIFIC ADJECTIVES (ITEMS 51–75)

Adjectives

#	Term (Adjective)	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
51	Empirical	प्रयोगात्मक (Prayogaatmak)	Based on direct observation or experiment rather than theory.	*"We gathered empirical evidence through twenty repeated trials."*
52	Exothermic	ऊष्माक्षेपी (Ooshmakshepi)	Releasing net heat energy into surrounding environment.	*"Dissolving sodium hydroxide in water is an exothermic reaction."*
53	Endothermic	ऊष्माशोषी (Ooshmashoshi)	Absorbing thermal energy from surroundings.	*"Ammonium nitrate dissolution creates an endothermic cold pack."*
54	Aqueous	जलीय (Jaleey)	Dissolved in, containing, or resembling liquid water.	*"Add aqueous copper sulfate solution to the test tube."*
55	Volatile	वाष्पशील (Vaashpasheel)	Evaporating rapidly at normal classroom temperatures.	*"Keep ether away from open flames because it is highly volatile."*
56	Viscous	चिपचिपा / श्यान (Shyaan)	Thick, sticky, and slow-flowing fluid consistency.	*"Motor oil becomes significantly less viscous as heat increases."*
57	Hygroscopic	आर्द्रताग्राही (Aardratagraahi)	Absorbing ambient moisture readily from the air.	*"Calcium chloride pellets are hygroscopic desiccants."*
58	Malleable	आघातवर्धनीय (Aaghaatvardhaneey)	Capable of being hammered or rolled into thin sheets.	*"Gold is exceptionally malleable and forms foil easily."*
59	Ductile	तन्य (Tanya)	Able to be drawn into thin elongated wires.	*"Copper is preferred for electrical cabling because it is ductile."*
60	Anaerobic	अवायवीय (Avaayaveey)	Living, acting, or occurring in absence of free oxygen.	*"Yeast produces ethanol during anaerobic fermentation."*
61	Aerobic	वायवीय (Vaayaveey)	Requiring free elementary oxygen for respiration or growth.	*"Aerobic respiration yields thirty-two ATP molecules per glucose."*
62	Insoluble	अघुलनशील (Aghulansheel)	Incapable of dissolving in a designated solvent.	*"Barium sulfate remains insoluble in concentrated acid."*
63	Soluble	घुलनशील (Ghulansheel)	Able to dissolve uniformly in a liquid liquid carrier.	*"Potassium nitrate is highly soluble in boiling water."*
64	Heterogeneous	विषमांगी (Vishamaangi)	Non-uniform composition with visible phase boundaries.	*"Oil and vinegar create a heterogeneous mixture when shaken."*
65	Homogeneous	समांगी (Samaangi)	Uniformly distributed composition and appearance throughout.	*"A well-stirred saline solution is completely homogeneous."*
66	Corrosive	संक्षारक (Sankshaarak)	Chemically eating away, eroding, or degrading materials.	*"Wear nitrile gloves when handling corrosive mineral acids."*
67	Radioactive	रेडियोधर्मी (Radiodharmi)	Spontaneously emitting ionizing nuclear particles or rays.	*"Store radioactive source materials inside thick lead cases."*
68	Saturated	संतृप्त (Santript)	Containing maximum possible dissolved solute concentration.	*"No more sugar dissolves in this saturated solution at room temp."*
69	Unsaturated	असंतृप्त (Asantript)	Able to dissolve additional solute at current temperature.	*"The solution is unsaturated, so added crystals dissolve readily."*
70	Supersaturated	अतिसंतृप्त (Atisantript)	Holding more solute than normal solubility limit.	*"Dropping a seed crystal into supersaturated liquid sparks crystallization."*
71	Permeable	पारगम्य (Paargamya)	Allowing fluids or gases to pass directly through pores.	*"Cell walls are permeable to small polar molecules."*
72	Impermeable	अपारगम्य (Apaargamya)	Not permitting passage of fluids or particles through material.	*"Clay soil forms an impermeable substrate that traps standing water."*
73	Biotic	जैविक (Jaivik)	Relating to or resulting from living ecological components.	*"Predators and soil bacteria are key biotic factors."*
74	Abiotic	अजैविक (Ajaivik)	Physical, non-living components of an ecological system.	*"Sunlight intensity, rainfall, and pH are abiotic variables."*
75	Isothermal	समतापी (Samataapi)	Occurring at a constant, unchanging temperature.	*"Gas expansion in this cylinder is an ideal isothermal process."*

PART 3: PRIMARY SCIENTIFIC VERBS (ITEMS 76–100)

Verbs: Action Words

#	Term (Verb)	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
76	Titrate	अनुमापन करना (Anumaapan Karna)	Measure solution volume to neutralize or reach target.	*"Titrate the hydrochloric acid against sodium hydroxide carefully."*
77	Decant	निथारना (Nithaarna)	Pour off supernatant liquid without disturbing sediment.	*"Decant the clear upper liquid into an Erlenmeyer flask."*
78	Calibrate	अंशांकन करना (Anshaankan Karna)	Adjust instrument against a verified measurement standard.	*"Calibrate the electronic balance using standard brass weights."*
79	Synthesize	संश्लेषण करना (Sanshleshan Karna)	Produce a chemical compound from precursor substances.	*"Students will synthesize aspirin from salicylic acid today."*
80	Precipitate	अवक्षेपित होना (Avakshepit Hona)	Form insoluble solid particles out of a solution.	*"Silver chloride will precipitate immediately upon mixing ions."*
81	Sterilize	रोगाणुरहित करना (Roganurahit)	Destroy all living micro-organisms and spores on tools.	*"Autoclave the culture glassware at 121°C to sterilize thoroughly."*
82	Inoculate	टीका लगाना / संरोपण (Sanropan)	Introduce bacteria or cells into a growth medium.	*"Inoculate the nutrient broth with a loopful of bacterial colony."*
83	Vaporize	वाष्पीकृत करना (Vaashpeekrit)	Convert solid or liquid substance into gaseous vapor state.	*"Liquid nitrogen vaporizes rapidly when exposed to air."*
84	Dissociate	वियोजित होना (Viyojit Hona)	Split into smaller constituent ions or molecules.	*"Strong mineral acids dissociate completely in aqueous solution."*
85	Condense	संघनन होना (Sanghanan Hona)	Transition from gaseous vapor state into liquid form.	*"Steam will condense on contact with the chilled condenser glass."*
86	Sublime	ऊर्ध्वपातन होना (Oordhvapatan)	Change directly from solid to vapor without melting.	*"Dry ice sublimes directly into gaseous carbon dioxide."*
87	Centrifuge	अपकेंद्रित करना (Apkendrit Karna)	Separate phases by rapid centrifugal rotational spinning.	*"Centrifuge the suspension to pellet the yeast cells at the bottom."*
88	Filter	छानना (Chhanna)	Pass fluid through porous medium to retain particulates.	*"Filter the turbid mixture using Whatman filter paper."*
89	Neutralize	उदासीन करना (Udaaseen Karna)	React acid and base to bring solution pH toward 7.	*"Neutralize the chemical spill with powdered sodium bicarbonate."*
90	Dissect	विच्छेदन करना (Vichhedan Karna)	Cut open biological specimen to inspect internal organs.	*"Dissect the flower to observe the stamen and carpel structure."*
91	Incubate	उष्मायन करना (Ooshmaayan Karna)	Maintain cells or bacteria under optimal growth heat.	*"Incubate the cultured plates at 37°C for twenty-four hours."*
92	Homogenize	समांगीकृत करना (Samaangeekrit)	Blend substances into a smooth, uniform phase emulsion.	*"Homogenize the leaf tissue in chilled buffer using a pestle."*
93	Oscillate	दोलन करना (Dolan Karna)	Move rhythmically back and forth about an equilibrium point.	*"The simple pendulum oscillates with an invariant time period."*
94	Corrode	संक्षारित होना (Sankshaarit Hona)	Deteriorate gradually due to chemical oxidative attack.	*"Acidic rain accelerates the rate at which iron bridges corrode."*
95	Fluoresce	प्रतिदीप्ति उत्पन्न करना (Pratideepti)	Emit visible light upon absorbing higher energy radiation.	*"Certain minerals fluoresce vivid green under ultraviolet lamps."*
96	Replicate	प्रतिकृति बनाना (Pratikriti Banana)	Duplicate exact copies of genetic material or an experiment.	*"Always replicate your lab trial three times to confirm reliability."*
97	Inhibit	अवरोधित करना (Avrodhit Karna)	Slow down, block, or prevent a biological or chemical pathway.	*"Heavy metal ions inhibit essential enzyme catalytic action."*
98	Accelerate	त्वरित करना (Tvarit Karna)	Increase rate of velocity change or reaction progress.	*"Adding a catalyst will accelerate the breakdown of peroxide."*
99	Decay	क्षय होना (Kshay Hona)	Decompose organically or undergo nuclear breakdown.	*"Uranium isotopes decay over geological epochs into lead."*
100	Coagulate	थक्का जमना (Thakka Jamna)	Change from liquid fluid to thickened gel or solid clot.	*"Heating milk proteins causes them to denature and coagulate."*

PART 4: SCIENTIFIC ADVERBS (ITEMS 101–125)

Adverbs

#	Term (Adverb)	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
101	Empirically	प्रयोगात्मक रूप से (Prayogaatmak)	Demonstrated through hands-on experimental trial.	*"Gas laws were validated empirically before molecular theory."*
102	Quantitatively	मात्रात्मक रूप से (Maatraatmak)	Expressed in measurable numerical values and units.	*"Measure the acid concentration quantitatively via titration."*
103	Qualitatively	गुणात्मक रूप से (Gunaatmak)	Described through observable non-numerical features.	*"Observe qualitatively whether the solution turns pink or turbid."*
104	Spontaneously	स्वतःस्फूर्त रूप से (Swatah-sphoort)	Occurring naturally without continuous external intervention.	*"The reaction occurs spontaneously under negative Gibbs free energy."*
105	Stoichiometrically	रससमीकरणमितीय रूप से	In exact proportions predicted by balanced equation.	*"Mix hydrogen and oxygen stoichiometrically in a 2:1 ratio."*
106	Reproducibly	पुनरुत्पादक रूप से (Punrutpaadak)	Yielding consistent results upon repeated trial execution.	*"The assay performs reproducibly across three independent labs."*
107	Endogenously	अंतर्जात रूप से (Antarjaat Roop Se)	Originating or produced internally from within an organism.	*"Insulin is synthesized endogenously by pancreatic beta cells."*
108	Exogenously	बहिर्जात रूप से (Bahirjaat Roop Se)	Introduced from external environmental sources outside body.	*"Apply antibiotics exogenously to inhibit bacterial lawn growth."*
109	Inversely	व्युत्क्रमानुपाती रूप से (Vyutkram)	Varying such that one factor falls as other rises.	*"Gas volume is inversely proportional to applied hydrostatic pressure."*
110	Directly	समानुपाती रूप से (Samaanupaati)	Varying in parallel constant multiplicative ratio.	*"Gas pressure varies directly with absolute kelvin temperature."*
111	Statistically	सांख्यिकीय रूप से (Saankhyikeey)	Confirmed through formal probability and p-value metrics.	*"The difference between treatment groups is statistically significant."*
112	Adiabatically	रुद्धोष्म रूप से (Ruddhoshm Roop Se)	Occurring without heat transfer in or out of system.	*"Compressing the gas adiabatically causes temperature to spike."*
113	Linearly	रैखिक रूप से (Raikhik Roop Se)	Scaling directly in a straight-line proportional trend.	*"Optical absorbance scales linearly with chemical concentration."*
114	Aseptically	रोगाणुरहित तरीके से (Aseptic)	Under conditions sterile from microbial contamination.	*"Transfer cell culture media aseptically under the laminar hood."*
115	Reversibly	उत्क्रमणीय रूप से (Utkramaneey)	Capable of returning to initial state upon path reversal.	*"The enzyme inhibitor binds reversibly at normal body temperature."*
116	Irreversibly	अनुत्क्रमणीय रूप से (Anutkramaneey)	Permanently changed with no possibility of restoration.	*"Extreme heat irreversibly denatures delicate protein enzymes."*
117	Homogeneously	समांगी रूप से (Samaangi Roop Se)	Dispersed uniformly throughout all spatial regions.	*"The dye diffuses homogeneously through the beaker of water."*
118	Simultaneously	एक साथ / समकालिक (Samkaalik)	Occurring or recorded at the exact same moment.	*"Record temperature and pressure simultaneously at thirty-second ticks."*
119	Uniformly	एकसमान रूप से (Eksamaan Roop Se)	Evenly distributed without localized variances or clumps.	*"Heat the bottom of the evaporating dish uniformly with the burner."*
120	Dropwise	बूंद-बूंद करके (Boond-boond Karke)	Adding a liquid reagent one single drop at a time.	*"Add hydrochloric acid dropwise while swirling the conical flask."*
121	Optically	प्रकाशिक रूप से (Prakaashik)	In relation to light transmission, reflection, or polarization.	*"The synthesized chiral compound is optically active in polarimetry."*
122	Vigorously	तेजी से / जोर-जोर से (Tezi Se)	With strong, forceful agitation or rapid speed.	*"Shake the separatory funnel vigorously, then vent internal vapor."*
123	Systematically	व्यवस्थित रूप से (Vyavasthit)	According to an organized, step-by-step rigorous method.	*"Test every suspect reagent systematically to isolate the contaminant."*
124	Radially	त्रिज्यीय रूप से (Trijyeey Roop Se)	Spreading outward straight from a central origin point.	*"Electric field lines emanate radially outward from a positive charge."*
125	Thermodynamically	ऊष्मागतिकीय रूप से (Ooshmagatiki)	Governed by energy states and thermodynamic feasibility.	*"The formation of diamond from graphite is thermodynamically unfavorable at standard pressure."*

PART 5: PRONOUNS, CONJUNCTIONS & INTERJECTIONS (ITEMS 126–150)

Structural Connectors

#	Grammar Category	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
126	Each (Pronoun)	प्रत्येक (Pratyek)	Every individual unit taken one by one in an experimental set.	*"Each of the test tubes contains ten milliliters of buffer."*
127	Either (Pronoun)	दोनों में से एक (Dono Mein Se Ek)	One or the other of two distinct options.	*"Either indicator will accurately show the equivalence point."*
128	Neither (Pronoun)	दोनों में से कोई नहीं	Not the first nor the second experimental condition.	*"Neither solution showed discoloration under ultraviolet light."*
129	Which (Pronoun)	जो कि / जिससे (Jo Ki)	Relative pronoun linking to an observation or consequence.	*"Add silver nitrate, which precipitates insoluble silver chloride."*
130	Both (Pronoun)	दोनों (Dono)	Referring to the two samples or conditions simultaneously.	*"Both cultures showed visible bacterial lawn formation."*
131	Whatever (Pronoun)	जो कुछ भी (Jo Kuchh Bhi)	Any quantity or element without restriction.	*"Whatever gas evolves is collected over water in the cylinder."*
132	None (Pronoun)	कोई भी नहीं (Koi Bhi Nahi)	Not one member of a sample set.	*"None of the controls displayed microbial contamination."*
133	All (Pronoun)	सभी (Sabhi)	The whole aggregate collection of samples or data points.	*"All trials produced consistent enthalpy measurements."*
134	Whichever (Pronoun)	जो कोई भी (Jo Koi Bhi)	Any selected alternative without bias.	*"Whichever reagent you add first, keep the temperature constant."*
135	Any (Pronoun)	कोई भी (Koi Bhi)	An unspecified sample from the entire population.	*"Did any precipitate form after chilling the solution?"*
136	Whereas (Conj.)	जबकि (Jabki)	In direct comparison with the contrasting scientific fact.	*"Metals conduct electricity, whereas non-metals act as insulators."*
137	Therefore (Conj.)	अतः / इसलिए (Atah / Isliye)	For that logical reason; drawing an evidence-based conclusion.	*"Litmus paper turned red; therefore, the solution is acidic."*
138	Provided that (Conj.)	बशर्ते कि (Basharte Ki)	Subject to the specified operational boundary or condition.	*"Enzyme rates peak, provided that temperature stays below 40°C."*
139	Hence (Conj.)	फलस्वरूप (Phalaswaroop)	As an immediate direct consequence of recorded data.	*"The density exceeds water; hence, the solid sinks to the bottom."*
140	Unless (Conj.)	जब तक कि... न (Jab Tak Ki)	Except in the explicit circumstance that.	*"No reaction occurs unless you heat the mixture with a flame."*
141	Whenever (Conj.)	जब कभी भी (Jab Kabhi Bhi)	At any point in time when an event takes place.	*"Whenever pressure rises rapidly, release the safety valve."*
142	Although (Conj.)	यद्यपि / हालांकि (Haalanki)	Even though; acknowledging an apparent anomaly.	*"Although the reaction is exothermic, it requires high activation energy."*
143	Inasmuch as (Conj.)	इस दृष्टि से कि (Is Drishti Se)	Since; considering that a scientific fact is true.	*"Inasmuch as mass is conserved, the total reactant mass equals product mass."*
144	Notwithstanding (Conj.)	के बावजूद (Ke Baavjood)	In spite of; despite the presence of contrary evidence.	*"The reaction proceeded, notwithstanding high ambient humidity."*
145	Lest (Conj.)	ऐसा न हो कि (Aisa Na Ho Ki)	To prevent an adverse or hazardous lab outcome.	*"Keep acids covered lest toxic vapors escape into the room."*
146	Observe! (Interj.)	ध्यान से देखिए! (Observe!)	Command instructing students to watch a visual physical change.	*"Observe! Tiny effervescent carbon dioxide bubbles are rising now."*
147	Caution! (Interj.)	सावधान! (Saavdhaan!)	Immediate safety warning alert regarding hazards.	*"Caution! Hot glassware looks identical to cold glassware."*
148	Eureka! (Interj.)	मिल गया! / खोज लिया!	Exclamation celebrating a sudden scientific discovery.	*"Eureka! The unknown crystalline mineral matches the diffraction spectrum."*
149	Halt! (Interj.)	रुकिए! / बंद करें!	Urgent verbal order to cease a lab operation immediately.	*"Halt! Shut down the vacuum pump before pressure implodes the flask."*
150	Look! (Interj.)	देखिए! (Dekhiye!)	Directing student attention to an emerging phenomenon.	*"Look! The indicator is undergoing a sharp pink-to-clear transition."*

PART 6: SPOKEN LABORATORY PHRASAL VERBS (ITEMS 151–175)

Phrasal Verbs: Set A

#	Phrasal Verb	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
151	Boil off	उबालकर उड़ा देना (Ubaalkar)	Evaporate completely by prolonged boiling.	*"Boil off the water to recover the dissolved salt crystals."*
152	Burn off	जलाकर नष्ट करना (Jalaakar)	Remove volatile organic impurities by high-heat combustion.	*"Burn off trace carbon residues in the porcelain crucible flame."*
153	Filter out	छानकर अलग करना (Chhaankar)	Separate solid particles from fluid via filtration.	*"Filter out the precipitate using qualitative filter paper."*
154	Settle down	नीचे बैठना (Neeche Baithna)	Sink to the bottom under gravity; form sediment.	*"Allow the dense suspension to settle down before decanting."*
155	Give off	उत्सर्जित करना (Utsarjit Karna)	Emit gas, odor, heat, or vapor during reaction.	*"Adding zinc to hydrochloric acid gives off hydrogen gas."*
156	Cool down	ठंडा करना (Thanda Karna)	Allow temperature to return to ambient room level.	*"Let the hot beaker cool down completely on the wire gauze."*
157	Heat up	गर्म करना (Garm Karna)	Raise the thermal temperature of a solution or system.	*"Heat up the mixture gently using a water bath rather than flame."*
158	Break down	विघटित होना (Vighatit Hona)	Decompose into simpler chemical constituents.	*"Digestive enzymes break down complex carbohydrates into glucose."*
159	Speed up	गति बढ़ाना (Gati Badhana)	Increase rate of an ongoing chemical or physical process.	*"Agitation will speed up the dissolution of solute crystals."*
160	Slow down	गति कम करना (Gati Kam Karna)	Retard or decrease the velocity of a reaction.	*"Lowering temperature will slow down bacterial metabolism."*
161	Carry out	संपन्न करना (Sampann Karna)	Conduct or perform an experiment systematically.	*"Carry out the extraction under a functional fume hood."*
162	Dry out	सुखाना (Sukhaana)	Remove all residual moisture or water traces completely.	*"Place the crystals in a desiccator to dry out overnight."*
163	Rinse out	धोकर साफ करना (Dhokar Saaf)	Wash glassware thoroughly with distilled water.	*"Rinse out the volumetric flask with deionized water before use."*
164	Pipette out	पिपेट से निकालना (Pipette Se)	Draw and dispense an aliquot using a pipette.	*"Pipette out exactly twenty-five milliliters of standard alkali."*
165	Scale up	पैमाना बढ़ाना (Paimaana Badhana)	Expand reaction volume from benchtop to pilot scale.	*"Scale up the batch reaction safely from grams to kilograms."*
166	Top up	भरकर पूरा करना (Bharkar Poora)	Add liquid until reaching the exact graduation line.	*"Top up the volumetric flask with distilled water to the etched mark."*
167	Shut down	बंद करना (Band Karna)	Power off lab apparatus safely after completion.	*"Shut down the spectrophotometer and close the gas main valve."*
168	Wipe off	पोंछकर साफ करना (Ponchhkar)	Clean liquid drops or residue from workbench or cuvette.	*"Wipe off fingerprints from the outside optical face of the cuvette."*
169	Zero out	शून्य पर सेट करना (Shunya Par)	Tare an electronic scale to cancel tare weight.	*"Zero out the analytical balance with the empty weighing boat on it."*
170	Set up	तैयार करना (Taiyaar Karna)	Assemble and arrange laboratory instruments for testing.	*"Set up the Bunsen burner, tripod, and wire gauze carefully."*
171	Take up	अवशोषित करना (Avshoshit Karna)	Absorb or absorb liquid or nutrient molecules.	*"Plant roots take up dissolved inorganic nitrogen from soil water."*
172	Light up	प्रदीप्त होना / जलाना (Pradipt)	Ignite flame or glow visibly under stimulation.	*"Light up the burner with the air hole partially closed."*
173	Drop off	गिरावट आना (Giraavat Aana)	Decline or diminish sharply in recorded value.	*"Enzyme efficiency drops off sharply once temperature exceeds 50°C."*
174	Wash away	बहाकर निकालना (Bahaakar)	Flush or remove unbound reagents with excess fluid.	*"Wash away unbound fluorescent antibodies with saline wash."*
175	Spin down	अपकेंद्रण से तलछट बनाना	Pellet suspended cells or debris using a centrifuge.	*"Spin down the bacterial harvest at 4000 RPM for ten minutes."*

PART 7: PHRASAL VERBS & LAB PHRASES (ITEMS 176–200)

Lab Phrases & Protocols

#	Term / Phrase	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
176	Plate out	प्लेट पर संवर्धन करना (Plate Par)	Spread diluted microbe culture onto agar surface.	*"Plate out the serial dilution onto nutrient agar using a spreader."*
177	Phase out	क्रमशः समाप्त करना (Kramashah)	Discontinue use of toxic reagent over a period.	*"The department decided to phase out mercury thermometers."*
178	Blow out	फूंक मारकर निकालना	Expel residual liquid droplet from a volumetric pipette.	*"Do not blow out the last drop; pipettes are calibrated to retain it."*
179	Check off	सूची में टिक करना (Tick Karna)	Mark off completed steps in a safety protocol checklist.	*"Check off every safety pre-condition before igniting the furnace."*
180	Stir in	हिलाकर मिलाना (Hilaakar)	Incorporate a chemical component gradually with stirring.	*"Stir in the crystalline solute continuously until fully dissolved."*
181	In vitro	कांच के पात्र में / प्रयोगशाला में	Conducted outside a living organism in a test tube.	*"The preliminary enzymatic screening was demonstrated in vitro."*
182	In vivo	जीवित शरीर में (Jeevit Shareer)	Studied within the context of a whole living organism.	*"The therapeutic drug showed promising metabolic results in vivo."*
183	In situ	मूल स्थान पर (Mool Sthaan Par)	Examined or monitored in its original natural place.	*"Sensor probes measure deep ocean water salinity in situ."*
184	Standard temperature & pressure	मानक ताप और दाब (STP)	Reference benchmark: 0°C (273.15 K) and 1 atmosphere.	*"One mole of ideal gas occupies 22.4 liters at standard temperature and pressure."*
185	Margin of error	त्रुटि की सीमा (Truti Ki Seema)	Statistical range accounting for measurement uncertainty.	*"The measured melting point falls well within the margin of error."*
186	Room temperature	कमरे का तापमान (Kamre Ka)	Standard ambient lab temperature (~20°C to 25°C).	*"Allow the autoclaved agar medium to cool down to room temperature."*
187	Serial dilution	क्रमिक तनुकरण (Kramik Tanukaran)	Stepwise stepwise dilution of a substance in solution.	*"Perform a ten-fold serial dilution to count bacterial colonies."*
188	Chain reaction	श्रृंखला अभिक्रिया (Shrinkhla)	Self-sustaining series of reactive events propagating forward.	*"Neutrons emitted during nuclear fission trigger a chain reaction."*
189	Fume hood	धूम्र कोष्ठिका (Dhoomra Koshthika)	Ventilated safety enclosure designed to vent toxic gases.	*"Always handle concentrated nitric acid inside a working fume hood."*
190	Limiting reagent	सीमांत अभिकर्मक (Seemant)	Reactant completely consumed first, capping product yield.	*"Calculate theoretical yield by identifying the limiting reagent."*
191	Boiling chip	उबाल रोधी टुकड़ा (Ubaal Rodhi)	Porous ceramic piece preventing superheating flash in distillation.	*"Add two boiling chips to the distillation flask to promote smooth boiling."*
192	Double-blind study	दोहरा-अंध अध्ययन (Dohra-Andh)	Trial where neither subjects nor testers know group assignments.	*"The vaccine efficacy was validated in a randomized double-blind study."*
193	Steady state	स्थिर अवस्था (Sthir Awastha)	Condition where dynamic inputs match outputs identically.	*"The continuous culture bioreactor achieved a steady state."*
194	Blank sample	रिक्त नमूना (Rikt Namuna)	Sample containing only solvent used to calibrate spectrophotometer zero.	*"Zero the spectrophotometer using a blank sample before reading."*
195	Supernatant liquid	अधिटेरी / उपरिप्लवी द्रव	The clear fluid overlying sediment after centrifugation.	*"Carefully pour off the supernatant liquid into a clean beaker."*
196	Flame test	ज्वाला परीक्षण (Jwaala Pareekshan)	Qualitative analysis identifying metal ions by characteristic flame color.	*"A persistent crimson flame test indicates the presence of strontium."*
197	Tollen's test	टॉलेन परीक्षण (Tollens Test)	Chemical assay using ammoniacal silver to detect aldehydes.	*"A shining silver mirror on the tube walls confirms Tollen's test."*
198	Half-life	अर्ध-आयु (Ardh-Aayu)	Time required for half of a radioactive nucleus sample to decay.	*"The radioisotope has a short half-life of roughly six hours."*
199	Buffer solution	बफर विलयन (Buffer Vilayan)	Solution resisting changes in pH upon minor acid or base addition.	*"Maintain enzyme assays in a phosphate buffer solution at pH 7.4."*
200	Water bath	जल ऊष्मक (Jal Ooshmak)	Laboratory vessel filled with heated water for uniform warmth.	*"Incubate the sensitive test tubes in a thermostatted water bath."*

PART 8: ACADEMIC SCIENTIFIC PHRASES (ITEMS 201–225)

Academic Writing Phrases

#	Scientific Phrase	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
201	In accordance with	के अनुसार (Ke Anusaar)	Fully conforming to established scientific laws or protocol.	*"Dispose of biohazard waste in accordance with biosafety laws."*
202	With respect to	के संदर्भ में / के सापेक्ष	Concerning or evaluated in relation to a specific parameter.	*"Plot rate of reaction with respect to reactant concentration."*
203	As a consequence of	के परिणामस्वरूप (Ke Parinaam)	Arising as a direct biological or physical effect of.	*"Cell lysis occurred as a consequence of extreme osmotic shock."*
204	Subject to	के अधीन (Ke Adheen)	Conditioned or affected by designated experimental parameters.	*"The protein folding stability is subject to ambient ionic strength."*
205	It is noteworthy that	यह ध्यान देने योग्य है कि	Directing reader attention to a prominent experimental observation.	*"It is noteworthy that no side products were detected in spectroscopy."*
206	Under the auspices of	के तत्वावधान में (Ke Tatvaavdhaan)	Conducted with backing or guidance of a scientific institution.	*"This clinical research proceeded under the auspices of the institute."*
207	In the presence of	की उपस्थिति में (Ki Upasthiti)	Occurring in the explicit chemical atmosphere or context of.	*"The substrate oxidizes readily only in the presence of copper ions."*
208	In the absence of	की अनुपस्थिति में (Ki Anupasthiti)	Under conditions lacking a specific reagent or enzyme factor.	*"In the absence of a catalyst, the reaction is imperceptibly slow."*
209	By virtue of	के आधार पर / के कारण	On account of an intrinsic physical or chemical property.	*"By virtue of its low density, helium ascends rapidly in air."*
210	Ceteris paribus	अन्य बातें समान रहने पर	Holding all other external variables unchanged and constant.	*"Ceteris paribus, increasing temperature expands gas volume."*
211	Mutatis mutandis	यथावश्यक परिवर्तनों सहित	Applying the same logic while making necessary minor changes.	*"The argument holds for chlorine, mutatis mutandis for bromine."*
212	On the basis of	के आधार पर (Ke Aadhaar Par)	Supported and reasoned by the gathered empirical findings.	*"We reject the hypothesis on the basis of recorded titration curves."*
213	To a large extent	काफी हद तक (Kaafi Had Tak)	Predominantly; governing most observed occurrences.	*"Enzyme denaturation is to a large extent irreversible once boiled."*
214	In broad terms	मोटे तौर पर (Mote Taur Par)	Explaining general scientific principle without minor caveats.	*"In broad terms, kinetic energy represents energy of motion."*
215	In terms of	के रूप में / के संदर्भ में	Expressed using specific measurement units or concepts.	*"Express the heat of combustion in terms of kilojoules per mole."*
216	Statistically significant	सांख्यिकीय रूप से सार्थक	Unlikely to have occurred merely by random experimental chance.	*"The experimental drug produced a statistically significant reduction."*
217	Rate-determining step	दर-निर्धारक पद (Dar-Nirdharak)	The slowest elementary step governing overall reaction speed.	*"Carbocation formation is the rate-determining step in SN1 substitution."*
218	For the purpose of	के उद्देश्य से (Ke Uddeshya Se)	Conducted to fulfill a defined experimental objective.	*"Add EDTA for the purpose of chelating free divalent cations."*
219	In close agreement	पूर्ण सहमति में (Poorn Sahmati)	Matching published theoretical expectations closely.	*"Our experimental lattice values are in close agreement with literature."*
220	Contrary to expectations	अपेक्षाओं के विपरीत (Apekshaon)	Yielding results contradicting initial theoretical predictions.	*"Contrary to expectations, adding solute lowered the boiling point."*
221	It is well established that	यह सर्वविदित है कि	Affirming a proven and universally accepted scientific fact.	*"It is well established that DNA stores hereditary genetic data."*
222	Under normal conditions	सामान्य परिस्थितियों में	In typical ambient terrestrial environmental conditions.	*"Under normal conditions, mercury exists as a heavy liquid metal."*
223	In line with	के अनुरूप (Ke Anuroop)	Consistent with earlier models or experimental literature.	*"The observed spectral shift is in line with the Doppler effect."*
224	With the exception of	को छोड़कर (Ko Chhodkar)	Excluding one specific chemical element or anomalous case.	*"All noble gases, with the exception of helium, possess an octet."*
225	Beyond the scope of	के दायरे से बाहर (Ke Daayre)	Outside the boundaries of the present inquiry or laboratory course.	*"Quantum relativistic corrections are beyond the scope of this unit."*

PART 9: SCIENTIFIC IDIOMS & METAPHORS (ITEMS 226–250)

Idioms & Sayings

#	Scientific Idiom	Hindi Meaning	Scientific Meaning	Spoken Lab / Writing Example
226	Light years ahead	बहुत आगे होना (Bahut Aage)	Far more advanced in technological or conceptual capability.	*"Crispr gene editing is light years ahead of older breeding techniques."*
227	Down to a science	पूरी तरह कुशल होना (Puri Tarah)	Mastered to the point of flawless, repeatable execution.	*"Our lab technician has the protein gel preparation down to a science."*
228	Acid test	कठिन परीक्षा / अग्निपरीक्षा	A decisive trial that conclusively proves value or truth.	*"A field trial under storm conditions was the acid test for the turbine."*
229	Quantum leap	अभूतपूर्व छलांग (Chhalaang)	A sudden, massive, and transformative breakthrough advancement.	*"Penicillin's discovery represented a quantum leap in infectious medicine."*
230	Litmus test	कसौटी (Kasauti)	A definitive single factor showing true nature or status.	*"Replication by independent peers is the true litmus test of good science."*
231	Blowing hot and cold	लगातार रुख बदलना (Rukh Badalna)	Showing fluctuating, erratic enthusiasm or inconsistent results.	*"The aging sensor has been blowing hot and cold with baseline readings."*
232	On the same wavelength	एक जैसी सोच होना (Ek Jaisi Soch)	Sharing identical understanding and conceptual perspectives.	*"The research team needs to be on the same wavelength regarding safety."*
233	Sparks fly	तीखी बहस होना (Teekhi Bahas)	Heated intellectual debate or emotional clash erupting.	*"Sparks fly during symposium debates over dark matter theories."*
234	Boiling point	चरम सीमा (Charam Seema)	Point of crisis where tensions or reactions turn volatile.	*"Pressure within the sealed autoclave is approaching its boiling point."*
235	Blink of an eye	पलक झपकते ही (Palak Jhapkat)	An extremely brief interval; virtually instantaneous.	*"Explosive decomposition of gunpowder happens in the blink of an eye."*
236	Tip of the iceberg	समस्या का छोटा सा हिस्सा	Only a tiny manifest fraction of a far deeper phenomenon.	*"Known bacterial species are only the tip of the iceberg in biology."*
237	Blind spot	अंध बिंदु / अनदेखी कमी	An area where someone lacks awareness or critical insight.	*"Overlooking instrumentation parallax error was the researcher's blind spot."*
238	Under the microscope	बारीकी से जांच के दायरे में	Subjected to rigorous, granular, and uncompromising examination.	*"The researcher's statistical claims are now under the microscope."*
239	Catalyst for change	परिवर्तन का सूत्रधार (Sootradhaar)	A factor that precipitates significant systemic transformation.	*"Publishing the ozone depletion findings was a catalyst for change."*
240	Chain reaction	श्रृंखला प्रतिक्रिया (Shrinkhla)	Cascade where one incident triggers a succession of consequences.	*"The oil spill triggered an ecological chain reaction across the coast."*
241	Meltdown	विनाशकारी विफलता / पिघलना	Disastrous collapse of control, containment, or composure.	*"Failure of the cooling pumps led directly to a nuclear reactor meltdown."*
242	Trial and error	प्रयास और त्रुटि (Prayaas Aur)	Discovering viable solutions by learning through repeated attempts.	*"Finding the ideal crystallizing solvent involved weeks of trial and error."*
243	Back to the drawing board	शुरुआत से नई योजना बनाना	Starting over completely after an experiment fails.	*"The catalyst synthesis failed; it is time to head back to the drawing board."*
244	Push the envelope	सीमाओं को आगे बढ़ाना (Seemaon)	Extend known boundaries of performance, speed, or knowledge.	*"Particle colliders push the envelope of high-energy physics."*
245	Reinvent the wheel	व्यर्थ मेहनत करना (Vyarth)	Waste effort recreating something that has already been perfected.	*"Use standard verified buffer tables; do not reinvent the wheel."*
246	Gauging interest	रुचि का आकलन करना (Ruchi Ka)	Assessing the degree of attention or receptivity to an idea.	*"The presenter paused after the demonstration to gauge interest."*
247	Silver bullet	रामबाण समाधान (Rambaan)	A simple, guaranteed universal panacea for a complex problem.	*"Scientists caution that carbon capture alone is not a silver bullet."*
248	In a vacuum	पृथक रूप से / अलगाव में	Considered in total isolation from practical context.	*"Scientific discoveries cannot be evaluated in a vacuum without ethics."*
249	Zero tolerance	शून्य सहनशीलता (Shunya Sahansheelata)	Policy of absolute, uncompromising refusal to allow errors.	*"The biosafety facility enforces zero tolerance for contamination."*
250	Blind testing	अज्ञात परीक्षण (Agyaat Pareekshan)	Testing where critical identifying labels are concealed to avoid bias.	*"Blind testing confirms that the colorimetric change is reliable."*